

Housing Management System - Testing & Bug Fix Documentation

Project: AT-BMS / HousingOS

Report Type: Testing, Bug Fixing, and Project Documentation

Language: Urdu/Hinglish

Status: In progress / based on reported issues and fixes discussed in chat

1. Project Overview

Housing Management System ek society/apartment management software hai. Is ka kaam residents, bills, complaints, documents, visitors, reports, aur property records ko digital tarike se manage karna hai.

Main Purpose

- Society ke daily operations ko manage karna.
- Residents aur admin ke liye ek centralized system dena.
- Billing, complaints, documents, reports, aur analytics ko ek jagah handle karna.

Main Users

- Admin
- Resident
- Tenant
- Security/Staff
- Committee members

2. Tech Stack

- **Frontend:** React + TanStack Start
- **Backend:** TanStack Start server functions
- **Database:** MySQL 8.0
- **Styling:** Tailwind CSS + component library
- **Language:** TypeScript
- **Containerization:** Docker / Docker Compose
- **Package manager:** Bun / npm

3. Project Flow

1. User login karta hai.
2. System session aur role check karta hai.
3. Dashboard open hota hai.
4. User module select karta hai: forms, property, analytics, documents, reports, etc.
5. Data database se load hota hai.
6. User actions database me save hoti hain.
7. Reports aur charts generate hotay hain.
8. Errors aane par console me logs milte hain.

4. Modules Identified

Core Modules

- Authentication / Session management
- Dashboard
- Forms catalog
- Property management
- Documents management
- Analytics
- Reports

Business Modules

- Residents
- Units / Buildings / Blocks
- Billing / Ledger
- Payments
- Complaints
- Maintenance
- Visitors / Security
- Notices / Forum / Community

Support Modules

- API helpers
- Auth helper
- Database schema
- Docker setup

5. Issues Found and Status

Bug 1: Dashboard date error

Screen: Dashboard (/)

Error: `TypeError: dateString.split is not a function`

Cause: Date value string nahi thi; sometimes Date object aa rahi thi.

Fix: Safe type checking add ki gayi.

Status: Fixed

Bug 2: Analytics page date handling

Screen: /analytics

Error: Same date split / date handling issue

Cause: created_at field par direct date conversion ho rahi thi.

Fix: String/Date dono cases ke liye safe handling add ki gayi.

Status: Fixed

Bug 3: Property page context error

Screen: /property

Error: `Context is not available`

Cause: Server-side context / auth helper usage me inconsistency thi.

Fix: Auth helper pattern standardize kiya gaya.

Status: Fixed

Bug 4: Reports API context error

Screen: Reports API

Error: `Context is not available`

Cause: Duplicate / inconsistent server context access.

Fix: Reports API ko standardized auth helper pattern pe shift kiya gaya.

Status: Fixed

Bug 5: Reports page UI overlap

Screen: /reports

Issue: Layout overlap ho raha tha

Cause: Container width aur spacing constraints missing the.

Fix: Responsive container, width limit, aur padding add ki gayi.

Status: Fixed

Bug 6: Forms page cards not clickable

Screen: /forms

Issue: Cards click par work nahi kar rahe the

Investigation: Code me Link wrapping already sahi tha.

Result: Ye false alarm nikla; route and links already properly configured the.

Status: No fix needed

Bug 7: Documents page not loading

Screen: /documents

Issue: This page didn't load

Possible Cause: Route/component runtime issue ya page ke andar data load fail ho raha hai.

Status: Open / needs final code-level verification

Bug 8: Analytics route/page load issue

Screen: /analytics

Issue: Page loading issue reported earlier

Status: Fixed after date handling corrections

Bug 9: Forms route click navigation issue

Screen: /forms

Issue: Cards click issue reported earlier

Status: Verified as working

6. Database and Environment Status

Docker / MySQL

- MySQL Docker container use kiya gaya.
- Port mapping 3308:3306 report hua.
- Database name: at_bms
- User: root
- Password: root123

Reported Connection Errors

- `connect ECONNREFUSED 127.0.0.1:3308`
- `getaddrinfo ENOTFOUND mysql`

Interpretation

- `ECONNREFUSED` ka matlab database host/port par available nahi tha.
- `ENOTFOUND mysql` ka matlab hostname resolve nahi ho raha tha, usually tab jab app Docker network me nahi hoti ya host galat use hota hai.

Suggested Correct Setup

- Agar app Docker ke andar chal rahi ho: `MYSQL_HOST=mysql, MYSQL_PORT=3306`
- Agar app host machine par chal rahi ho: `MYSQL_HOST=127.0.0.1, MYSQL_PORT=3308`

7. Testing Checklist

Tested Areas

- Login flow
- Dashboard data rendering
- Forms page navigation
- Analytics calculations
- Property page loading
- Documents page loading
- Reports page layout
- Database connectivity

Expected Results

- Pages load correctly
- No runtime crash
- Dates properly display hon
- API context errors na aayen
- UI overlap na ho
- Database connection stable ho

8. Fix Summary

Fixed Items

- Dashboard date handling
- Analytics date handling
- Property API context usage
- Reports API context usage
- Reports page layout overlap

Verified Working

- Forms page cards navigation
- Documents API pattern consistency
- Database Docker setup concept

Still Needs Verification

- /documents page loading issue
- Final browser-level re-test after fixes

9. Recommended Next Steps

1. Documents page ka full code check karo.
2. Browser console me exact stack trace nikaalo.
3. Route file aur loader/server function verify karo.
4. Final manual testing run karo.
5. README / setup guide ko project me add karo.
6. API docs aur bug log maintain karo.

10. Manager Report Summary

Project Status: Mostly fixed and structured

Working Areas: Dashboard, analytics, property, reports, forms navigation

Main Remaining Issue: Documents page load issue

Overall: System ke major bugs resolve ho gaye hain, aur project documentation aur final verification ke stage par hai.

11. Conclusion

Yeh Housing Management System ek detailed society management platform hai. Is me billing, complaints, property, documents, reports, aur analytics jese modules hain. Ab tak jo major runtime bugs report hue the un me se bohat se fix ho chuke hain. Final step me /documents page ko verify karna aur documentation ko project repository me save karna chahiye.